AP-BidIQ — Product Requirements Document (v1, Hackathon Demo)

Owner: Mohineesh / Cittaai **Last updated:** 2026-04-28 **Status:** Locked for 2-day demo build **Hackathon:** AI-Based Bid Document Drafting and Evaluation Automation — AP Govt. Infrastructure & Investment Department **Demo deadline:** 2026-04-30 (T+2 days from 2026-04-28)

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1. Problem

Public procurement in Andhra Pradesh runs on long, structured tender packages (the EPCC Fishing Harbours Phase II tender we received is 16 sections, ~1 MB of text, 19 vendor forms). Today:

- **Drafting** RFPs is manual copy-paste from prior tenders. Project-specific thresholds and corrigenda are easy to miss.
- **Validation** of submitted bids is a checklist exercise done by hand. Mandatory annexures are missed, BG (Bank Guarantee) names don't match the JV name, bill totals don't reconcile, and stale solvency certificates slip through.
- **Evaluation** at qualification stage is binary (pass/fail) but spread across ~15 mandatory items + computable thresholds (turnover, bid capacity formula, specialized-work quantities). Errors here trigger rejections, appeals, and project delays.
- **Corrigenda** mid-tender (e.g., the live Corrigendum-1 in our corpus changed bid-capacity lookback 5→10 yrs) are tracked manually and inconsistently applied.

A procurement officer evaluating 5 bids spends ~2–3 days per tender on validation alone. Subjectivity and inconsistency cost both speed and trust.

2. Opportunity

A focused AI tool that ingests an RFP package + corrigenda → versions the active rules → ingests vendor bids → produces a **deterministic compliance scorecard with cited sources** that mirrors the human Evaluation Statement format. Officer reviews and approves; AI never auto-rejects.

This directly hits the hackathon brief's PoC success criteria: - ≥85% accuracy on missing-clause / compliance-gap detection - ≥90% match between AI verdict and human expert verdict

3. Goals (v1, 2-day scope)

G1. Ingest the supplied EPCC Fishing Harbours Phase II tender package + Corrigendum-1 → produce a versioned, machine-readable "active rules" object (thresholds, mandatory clauses, qualification criteria).

G2. Ingest a vendor bid (DOCX/PDF) → extract structured data for the 8 highest-value forms (Form-3, 4, 5A, 5B, 5C, 6A, 14, 19).

G3. Run a three-layer validator (deterministic rules → LLM clause semantics → cross-bid anomaly) and produce a per-bid compliance scorecard with **citations to source clauses**.

G4. Generate an Evaluation Statement output that mirrors the human evaluator template in the corpus.

G5. Demonstrate ≥85% defect detection on a benchmark set of 5 synthetic bids (1 clean + 4 with planted defects).

G6. Ship a live, deployed web app the panel can click through.

4. Non-Goals (explicit cuts for v1)

- **Drafting assistant.** Brief asks for it; v1 does evaluation only. Demo will mention drafting as a roadmap module, not a built feature.
- **Multi-department / multi-tender-type generalization.** v1 is scoped to **one tender type** (EPCC Fishing Harbour) using **the supplied corpus only**.
- **Real e-procurement integration.** No live API to AP's e-procurement portal. Mocked.
- **Telugu / bilingual UI.** Corpus is English-only; English-only UI for v1.
- **Auth / RBAC.** Single-officer demo; no login screen. Mention as roadmap.
- **Real OCR for scanned PDFs.** v2: shipped Tesseract + poppler pipeline with English + Telugu language packs. Toggle in upload UI.
- **Drafting assistant clause-suggestion RAG.** Not in v1.
- **Production audit/security hardening.** Audit log is implemented (it's a brief requirement) but not pen-tested.

5. Users & Personas

Primary user — Procurement Officer (Department / RTGS): - Reviews 3–10 bids per tender at qualification stage. - Wants: fast, defensible verdicts with traceable sources. - Pain: manual checklist work, hard-to-defend subjective calls, missed corrigenda.

Secondary user — Department Head (read-only): - Wants dashboards and audit confidence; not in v1 UI but mentioned in pitch.

Stakeholder — Vendor (not a user of v1): - Benefits indirectly from consistent, transparent evaluation.

6. Demo target & success criteria

#	Criterion	Target	How measured
S1	End-to-end demo runs live on deployed URL	Yes	Judge clicks → working app
S2	Versioned RFP rules generated from corpus + Corrigendum-1	Yes	UI shows pre/post-corrigendum diff for bid-capacity lookback
S3	Vendor bid ingest produces structured JSON	8 forms	Inspector view in UI
S4	Defect detection accuracy on synthetic benchmark	≥85%	Benchmark report page in app
S5	Every finding has a clickable citation	100%	Every red/green item links to source clause text
S6	Evaluation Statement output matches human template structure	Side-by-side OK	Compare to corpus's Evaluation Statements doc
S7	Audit log is append-only and tamper-evident	Hash-chained	Show audit log view with hash per row

7. Functional requirements

7.1 RFP Ingest (FR-RFP)

- **FR-RFP-1**: System accepts a folder of RFP section files (DOCX/DOC/PDF) and one or more corrigendum files.
- **FR-RFP-2**: System parses each section into a structured `Tender` object: `{tender_id, title, department, sections[], thresholds[], mandatory_clauses[], forms_required[], qualification_criteria[]}`.
- **FR-RFP-3**: System parses each corrigendum into a list of `Patch` operations (clause replace, threshold update, deadline extension).
- **FR-RFP-4**: System produces a versioned `ActiveRules` view: original + N corrigenda applied = current effective rules. Each rule carries its source provenance (section + clause + which corrigendum, if any, modified it).
- **FR-RFP-5**: UI shows a diff view: "What changed because of Corrigendum-1?"

7.2 Vendor Bid Ingest (FR-BID)

- **FR-BID-1**: System accepts a vendor bid as a single DOCX or PDF (or a folder).
- **FR-BID-2**: For each of the 8 priority forms (3, 4, 5A, 5B, 5C, 6A, 14, 19), system extracts a typed JSON record per the form's schema.

- **FR-BID-3**: System computes derived fields: 3-year average turnover, total similar-work value, available bid capacity (`A × N × 3 − B`), JV partners list, signed-form list.
- **FR-BID-4**: System stores raw bid + extracted JSON + extraction confidence per field.

7.3 Validator (FR-VAL)

Layer 1 — Deterministic checks (must implement all 10): - **FR-VAL-1.1**: All mandatory forms present (Form-19 checklist completeness). - **FR-VAL-1.2**: EMD/Bid Bond present, amount ≥ TDS-specified threshold, validity ≥ 180 days from bid submission date. - **FR-VAL-1.3**: BG issuer's named bidder matches Form-14 JV entity name (string match + fuzzy). - **FR-VAL-1.4**: Solvency certificate dated within 6 months of bid date. - **FR-VAL-1.5**: Bid capacity: `A × N × 3 − B ≥ Bid Value`. - **FR-VAL-1.6**: Annual turnover (best of last 7 years) ≥ TDS threshold. - **FR-VAL-1.7**: Similar-work value (best of last 5 years) ≥ TDS threshold. - **FR-VAL-1.8**: Specialized work thresholds (breakwater Rmt, dredging Cum, piling diameter & Rmt) all met. - **FR-VAL-1.9**: Bill summation: Σ(Bills 1–5) = Grand Summary total (tolerance ≤ 0.1%). - **FR-VAL-1.10**: Performance Security validity covers contract end date + DLP + 90 days.

Layer 2 — LLM clause semantics (OpenAI GPT-4o, must cite source, PII redacted): - **FR-VAL-2.1**: Integrity Pact present and signed by authorized signatory listed in Form-2 PoA. - **FR-VAL-2.2**: JV agreement contains explicit "joint and several liability" language. - **FR-VAL-2.3**: Power of Attorney scope covers bid signing + contract execution. - **FR-VAL-2.4**: Form-12 declarations: no deviations, no intermediaries, no exceptions. - **FR-VAL-2.5**: Each finding returns a citation `{section, clause, page_or_offset, quoted_text}`.

Layer 3 — Cross-bid anomalies (run when ≥2 bids loaded): - **FR-VAL-3.1**: Fuzzy-match experience claims (project name + client + value + dates) across all bids; flag ≥80% similarity. - **FR-VAL-3.2**: Subcontractor name overlap across bids. - **FR-VAL-3.3**: Methodology language similarity (embedding cosine ≥ 0.92).

7.4 Evaluation Statement (FR-EVAL)

- **FR-EVAL-1**: Generate a per-bid statement matching the column structure of the human "Evaluation Statements Updated.docx" in the corpus (criteria | required | submitted | meets/doesn't | remarks).
- **FR-EVAL-2**: Roll up to a single qualification verdict: **Qualified / Not Qualified**, with reason list.
- **FR-EVAL-3**: Export as DOCX (matches human format) and JSON.

7.5 Officer Dashboard (FR-UI)

- **FR-UI-1**: Tender list page → click a tender → see active rules + corrigendum diff.
- **FR-UI-2**: Bid upload page (drag-drop DOCX/PDF).
- **FR-UI-3**: Bid detail page: scorecard with red/yellow/green per check, citation drawer that opens to show source clause.
- **FR-UI-4**: Tender comparison page: all bids side-by-side with cross-bid anomaly flags.
- **FR-UI-5**: Audit log page: append-only, hash-chained, every AI finding + (placeholder) officer action.

7.6 Audit Log (FR-AUD)

- **FR-AUD-1**: Every validator finding writes one row: `{ts, tender_id, bid_id, check_id, verdict, evidence_ref, model_version, prev_hash, this_hash}`.
- **FR-AUD-2**: `this_hash = SHA256(prev_hash || row_payload)`.
- **FR-AUD-3**: UI shows a "verify chain" button.

8. Non-functional requirements

	Requirement	Target
NFR-1	Time to validate one bid (end-to-end)	< 60 s
NFR-2	Concurrent bids in benchmark run	≥ 5
NFR-3	LLM cost per bid validation	< ₹50 (~$0.60) — use prompt caching on RFP context
NFR-4	Citation precision (citation actually points to relevant text)	≥ 90% on benchmark
NFR-5	Deployment: live URL	Vercel + Railway + Neon
NFR-6	Source code license	MIT in repo

9. System architecture

```
┌─────────────────────────┐         ┌─────────────────────────┐
│  Next.js (Vercel)       │  HTTPS  │  FastAPI (Railway)      │
│  • Officer dashboard    │ ──────► │  • /tenders, /bids,     │
│  • Upload UI            │         │    /validate, /eval     │
│  • Citation drawer      │         │  • Claude API client    │
│  • Audit log viewer     │         │  • Validator layers     │
└─────────────────────────┘         └────────────┬────────────┘
                                                  │
                                                  ▼
                                      ┌─────────────────────┐
                                      │  Postgres (Neon)    │
                                      │  + pg_trgm for fuzzy│
                                      │    matching         │
                                      └─────────────────────┘
                                                  │
                                                  ▼
                                      ┌─────────────────────┐
                                      │  Object storage     │
                                      │  (Railway volume or │
                                      │   S3) for raw docs  │
                                      └─────────────────────┘

External: Anthropic Claude API (Sonnet 4.6 for validation, Haiku 4.5 for batch
extraction). Prompt caching on RFP context — single 200K-token cache hit
serves all bids for that tender.
```

10. Data model (Postgres)

```
tenders          (id, title, department, created_at)
sections         (id, tender_id, name, raw_text, source_path)
corrigenda       (id, tender_id, name, raw_text, applied_at)
patches          (id, corrigendum_id, target_clause, op, before, after)
active_rules     (id, tender_id, version, json_blob, generated_at)
forms_required   (id, tender_id, form_no, title, schema_json)
mandatory_clauses(id, tender_id, code, title, source_section, source_text)

bids             (id, tender_id, vendor_name, jv_partners[], submitted_at, raw_path)
form_extractions (id, bid_id, form_no, json_payload, confidence, raw_excerpt)

validations      (id, bid_id, check_id, layer, verdict, severity, message,
                  citation_section, citation_clause, citation_text, model_version)

cross_bid_flags  (id, tender_id, bid_a, bid_b, flag_type, similarity, evidence)

eval_statements  (id, bid_id, verdict, reasons_json, docx_path, json_path)

audit_log        (id, ts, actor, action, payload_json, prev_hash, this_hash)
```

11. Demo dataset

Real corpus (in `data/corpus/`): - EPCC Fishing Harbours Phase II — 13 sections (Sections 1-10, 6A/B/C) - Top-level RFP for PMC services - Evaluation Statements Updated.docx (human evaluator template — our ground truth) - Corrigendum-1.docx (live amendment to demo versioning)

Synthetic vendor bids (`data/synthetic-bids/`): 1. `bid-01-clean.docx` — fully compliant, should pass. 2. `bid-02-missing-bg.docx` — Form-19 ticked but BG annexure absent. Tests FR-VAL-1.1 + 1.2. 3. `bid-03-bg-name-mismatch.docx` — BG issued to lead JV partner, not the JV entity. Tests FR-VAL-1.3. 4. `bid-04-capacity-shortfall.docx` — bid capacity computed as `A × N × 3 − B` falls below bid value. Tests FR-VAL-1.5 + arithmetic. 5. `bid-05-stale-solvency-and-bill-mismatch.docx` — solvency cert 9 months old + Bills don't sum to Grand Summary. Tests FR-VAL-1.4 + 1.9. Plus copies methodology language verbatim from bid-01 to trigger FR-VAL-3.3.

Ground-truth labels in `data/synthetic-bids/labels.json`:

```
{
  "bid-01-clean": { "verdict": "Qualified", "expected_findings": [] },
  "bid-02-missing-bg": { "verdict": "Not Qualified", "expected_findings": ["FR-VAL-1.1", "FR-VAL-1.2"] },
  ...
}
```

Accuracy = (correct verdicts + correct findings) / total expected.

12. UI flows

```
/                            → Tender list (1 tender pre-loaded)
/tenders/:id                 → Active rules view + corrigendum diff
/tenders/:id/bids            → Bid list (5 synthetic bids pre-loaded)
/tenders/:id/bids/upload     → Drag-drop new bid
/bids/:id                    → Scorecard (3 layers, citations)
/bids/:id/eval-statement     → Generated Evaluation Statement (DOCX preview)
/tenders/:id/compare         → Cross-bid anomaly view
/audit                       → Audit log with hash-chain verifier
/benchmark                   → Accuracy report on synthetic set (for the panel)
```

13. Out of scope (roadmap to mention on stage)

- Drafting assistant (RAG over clause library)
- Multi-tender-type templates (works, services, goods, consultancy)
- Telugu UI + bilingual clause search
- Real e-procurement portal integration
- Vendor portal (today's UI is officer-side only)

- Auth / RBAC / SSO with department IdP
- OCR for scanned PDFs
- Tender drafting assistant
- Mobile-responsive UI
- Real anonymization pipeline at ingest

14. Risks & mitigations

Risk	Mitigation
Claude API key not ready in time	Scaffold + run extractor against text-extracted corpus; switch to API once key arrives
2-day timeline slips	Drop FR-VAL-3 (cross-bid) first, then FR-AUD hash chain, then UI polish — never the deterministic validator
Deploy fails day-of	Local-laptop fallback ready; record screen video as last-resort backup
Synthetic bids feel toy-ish to judges	Frame on stage: "we used your real RFP and engineered defects mirroring patterns in past tender appeals"
Judge asks "what about scanned PDFs?"	Roadmap answer; mention Claude vision + Tesseract fallback
LLM gives wrong citation	Citation is rendered with quoted text; officer sees the quote and can verify in one glance

15. Open questions

1. Do we have a domain SME (retired procurement officer / consultant) we can show this to before stage? Strongly recommended for sanity-check.
2. Is presentation on AP-provided hardware or our laptop? Affects deploy vs. local fallback priority.
3. Are we expected to leave the codebase with AP afterward? If yes, add a 1-page handoff doc to docs/.
4. Any AP branding constraints (colors, logo) to bake into UI?

Appendix A — Mandatory check matrix (all 15 from corpus, marked v1 vs roadmap)

#	Check	Source	v1 layer
1	EMD present + valid + correct amount	ITT 1.12, Form-B	L1
2	Tender Document Integrity (Form-12)	Form-12	L2
3	Company Registration	ITT 1.6.2(a)	L1
4	EPF/ESI Registration	ITT 1.6.2(b)	L1
5	PAN & GST Registration	ITT 1.6.2(c)	L1
6	Power of Attorney (notarized, ≤6mo)	Form-2	L1 + L2
7	Audited financial statements (3 yrs)	Form-4, QR 3.1	L1
8	Experience certificates (gov: EE+SE; pvt: TDS+CA)	Form-5A/B/C, QR 4.1	L2
9	Equipment list	Form-8	L1 (presence only)
10	Key personnel + CVs	Form-9	L1 (presence only)
11	Integrity Pact signed	Form-13	L2
12	JV Agreement (joint & several)	Form-14	L2
13	PoA for Lead JV Member	Form-15	L2
14	No-exceptions declaration	Form-12	L2
15	Bid Form + Price Schedule arithmetic	Section 4 + 10	L1

Appendix B — Bid capacity formula

Per ITT 1.6.1 (as amended by Corrigendum-1):

```
Available Bid Capacity = (A × N × 3) − B

A = max annual contract value executed in any year of the last 10 years
    (was 5 years pre-corrigendum)
N = duration of current contract in years
B = value of existing commitments and ongoing works to be completed in
    next N years
```

Pass condition: `Available Bid Capacity ≥ Submitted Bid Value`.